Epidemiology of hip and groin injuries in sport: a systematic review with meta-analysis and meta-regression of 5914 injuries from over 9 million exposure hours

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ABSTRACT

Objective To systematically review hip and groin injury (HGI) incidence, burden, prevalence and the proportion of HGI relative to all injuries in competitive sports. A secondary aim was to determine the impact of potential effect moderators on the incidence of HGI in athletes.
Design Systematic review and meta-analysis.
Data sources MEDLINE, Web of Science, SPORTDiscus, PEDro, Embase and Scopus.
Eligibility criteria Prospective studies reporting a minimum of one epidemiological measure: HGI injury incidence, burden, prevalence or proportion of all injuries. Reporting quality was assessed using the checklist of issues that should be included in reports of football injury studies.
Results 71 studies (85% football studies) comprising 5914 injuries across 9 441 381 exposure hours were included. HGI incidence was 0.71/1000 hours, accounting for 11% of all injuries, with 6% affecting the hip adductors (0.41/1000 hours). Injury burden was 11.5 days per 1000 hours, while seasonal prevalence was 19%. HGI incidence rate increased with higher average cohort age (p=0.046). The incidence rate was 1.8 times higher in male compared with female athletes (95% CI 1.25 to 2.7, p=0.004), and 6.5 times higher during matches compared with training sessions (95% CI 3.9 to 10.9, p<0.001). Injury incidence was not significantly moderated by average cohort height (p=0.87), body mass (p=0.52), body mass index (p=0.58), year of data collection (p=0.13) or playing level (p=0.89).
Conclusions The incidence of HGI is 0.7 per 1000 hours of exposure, comprising 11% of all sporting injuries. Incidence rates increase with older age, and males are at higher risk of injury compared with females. The high proportion of HGI warrants efforts to reduce the occurrence of groin injuries in sport.
PROSPERO registration number CRD42024573833.

WHAT IS ALREADY KNOWN

⇒ Hip and groin problems are among the most common injuries in multidirectional sports, such as football, rugby or ice hockey.
⇒ A common terminology around groin injuries has been established in the last decade through the Doha agreement.
⇒ Different clinical entities can cause a groin injury.

WHAT ARE THE NEW FINDINGS

⇒ This dataset is the most extensive collection of hip and groin injuries, providing valuable insights into the outcomes presented.
⇒ The incidence of hip and groin injuries is 0.71 per 1000 exposure hours, while the injury burden is 11.5 lost days per 1000 hours.
⇒ Adductor-related injuries are the most common clinical entity and represent 6% of all injuries.
⇒ The incidence of hip and groin injuries is moderated by non-modifiable factors, with matches, male sex and older age associated with a greater injury incidence rate.

INTRODUCTION

Hip and groin injuries (HGIs) are a complex problem in sports that involve rapid acceleration and deceleration, sprinting and sudden changes of direction,[1] such as soccer,[2 3] Gaelic football,[4] rugby[5] and hockey.[6 7] Injuries negatively impact player availability and team performance, while posing a financial burden to the clubs.[8] Although most groin injuries affect the hip adductor muscles,[9 10] different clinical entities, such as iliopsoas-related,[11] inguinal-related,[12] pubic-related[13] or hip-related conditions,[14] can cause pain in the groin region according to the Doha agreement on groin pain terminology and definitions.[15–18] This issue is larger when cases that do not lead to time loss are also taken into consideration, since many groin injuries are the result of overuse, leading to pain and/or reduced performance without activity cessation (ie, non-time-loss injury). In fact, considering only groin injury time-loss cases underestimates the true injury prevalence by 70% to 90%.[19 20] Minimising the risk of injury is of utmost importance to maximise the benefits of sporting participation. The injury prevention sequence consists of four steps: presentation of epidemiological data, identification of injury mechanisms and risk factors, and the introduction and testing of prevention strategies. Understanding the epidemiology of injuries (ie, incidence, prevalence and burden) is the critical first step.[21]

Understanding factors that influence the incidence of HGI is also beneficial. Variables such as the playing scenario (eg, match vs training) and athlete characteristics (eg, sex, age and body mass) have been associated with HGI risk.[22 23] Exploring

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these factors can help monitor at-risk athletes (ie, secondary prevention) and tailor individualised programmes to prevent reinjuries.[24] However, focusing solely on count data (ie, incidence, prevalence) overlooks injury severity; injuries that require long recovery periods may demand equal or greater effort to prevent.[25] Analysing injury incidence while considering injury burden improves the understanding of which injuries necessitate further efforts to reduce injury rates.[26]

Numerous prospective injury surveillance studies across sports have been published, most focusing on general injury incidence without providing entity-specific data or other epidemiological measures (eg, injury burden) recommended by the International Olympic Committee.[27] Additionally, no meta-analysis has been published in this field to date. Reporting epidemiological data according to clinical entities and performing a quantitative synthesis will improve certainty around the obtained results, facilitating diagnosis and improving the design and implementation of preventive programmes. Thus, the primary aim of this study was to determine the injury incidence, burden, the proportion of which all injuries are composed of HGI and prevalence of groin injuries in sports. A secondary aim was to determine the impact of potential effect moderators on the incidence of groin injuries in athletes.

METHODS

This systematic review adheres to Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. It was prospectively registered in the International Prospective Register of Systematic Reviews (PROSPERO: CRD42024573833) in August 2024. A deviation from the original protocol occurred in relation to the secondary analyses, as sensitivity analyses examining the potential influence of studies with lower reporting quality on the overall estimates were not prespecified. This deviation was agreed on by the research team before data analysis commenced.

Literature search strategy

Studies were identified through a systematic search on six databases (MEDLINE, Web of Science, SPORTDiscus, PEDro, Embase and Scopus), from inception to 14 September 2024. In addition, previous systematic reviews and reference lists of included studies were also checked. Details of the key terms used for database search are shown in online supplemental table S1. The search terms were determined to align with the aims of the review. Obtained records were imported into Zotero (V.7.0.5) and duplicates were removed.

Study selection

Two authors (MQ-C and IC) independently screened titles and abstracts against eligibility criteria, before screening full texts of relevant studies for final inclusion. Disagreement between the two reviewers was solved by discussion, and a third reviewer (HO) was involved if consensus was not achieved. Studies were included when they reported a minimum of one of these outcomes: injury incidence, injury burden, injury prevalence, proportion of groin injuries in relation to all injuries, or presented sufficient information to calculate one outcome measure. Studies should follow a prospective design and include participants involved in competitive sports, with no restrictions on sex, age, competitive level or sport. Clinical trials with a control group which had not undergone any intervention were also included. Additionally, studies were required to be written in English and published in a peer-reviewed journal. Grey literature (ie, conference papers, abstracts, theses and unpublished reports) was not considered for inclusion.

Reporting quality

We are unaware of any tools available to accurately assess the risk of bias in prospective injury surveillance studies. Furthermore, an educational review by Büttner *et al* recommended against modifying existing risk of bias tools by adding or omitting items to suit specific study characteristics.[28 29] Therefore, an assessment of risk of bias for the included studies was not performed. Instead, all studies were mapped to the methodological domains of the 'Checklist of issues that should be included in reports of studies of football injuries' to evaluate reporting quality, following the approach used in a previous systematic review.[30] Reporting quality refers to the extent to which an original research article provides complete and transparent information regarding the design, conduct, analysis and results of a study.[29] Reporting quality for the included articles was performed by two reviewers (MdVS and IC) with conflicts resolved by consensus. The description of each item is available in online supplemental table S2.

Data extraction

Data were extracted by one author (MQ-C) using a customised Excel spreadsheet. Extracted data pertaining to the key outcome measures included the total number of groin injuries, total exposure hours, total number of injured participants, lay-off days due to injury, number of participants at risk and total number of injuries (of any type), if available. In some cases, required data were not directly reported in the relevant articles, but were calculable from other data reported within the article. Demographic characteristics (age, height, body mass, sex, sport and level of competition) were extracted, as were methodological details including injury definition (ie, time-loss injury, medical attention injury, all complaints), the year of observation and details of exposure scenario (eg, training vs competition). Where data were presented only graphically, we extracted data using WebPlotDigitizer (V.5.2).

Data analysis

Meta-analysis and meta-regression were conducted using the 'metafor' package in R (The R Foundation for Statistical Computing).[31] Four different meta-analyses were performed for each of the primary outcomes: (1) injury incidence rate per 1000 exposure hours, (2) injury burden per 1000 exposure hours, (3) proportion of groin injuries relative to all injuries and (4) prevalence of HGI. For each meta-analysis conducted, data were pooled across studies using a random effects model, with a restricted maximum likelihood method used to estimate variance. Note that we used a multilevel model to account for dependent effect sizes and imputed covariance matrices (assuming correlations of 0.7 and 0.5 for the same and different injury subtypes, respectively) to account for dependent standard errors, where appropriate. Since these correlation values were estimated, we employed cluster-robust inference methods with a small sample or 'CR1' adjustment.[32] As prevalence was expected to be sensitive to the analysis period used by each study, we also included the prevalence period (in months) as a moderator for this model. The outcome variables for each study were transformed prior to pooling, owing to the more favourable properties of the transformed data for meta-analysis, but were back transformed prior to reporting for ease of interpretation. For incidence and burden, the transformation applied was the natural log, while

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prevalence and hip/groin injury as a proportion of all injuries used the arcsine transformation.[33] Along with the 95% CI, we derived a 95% prediction interval (PI) to describe the range of true effects expected in 95% of future studies on this field.[34] Additionally, we report the number of effect sizes (k), injuries and exposure hours (if applicable) included in each meta-analytical model. Where exposure was 'athletic exposures', these were excluded owing to being a different metric to exposure hours and very low number of studies using this definition. When a study presented both 'time-loss' and 'all' injuries, only the 'all' effect sizes were included. When inspection of funnel plots and modified Egger's regression test evidenced the presence of small study bias or selective reporting, we used a two-step approach to re-estimate the pooled meta-analytic result as a sensitivity analysis to determine the effect on our inferences.[35] Meta-regression was performed for continuous (age, body mass, height, sex proportion, body mass index (BMI) and year of data collection) and categorical (injury scenario (ie, match or training session) and competitive level (ie, elite or amateur population)) variables to account for potential moderator effects on the incidence rates. Since body mass, BMI and stature could be expected to vary due to age (especially in younger cohorts), we performed post hoc analysis for these models where age was included as a covariate. We report the percentage change for each one-unit increase in the moderator, beta (β) coefficient (which represents the change in log incidence for a 1 unit increase in the continuous moderator, or from one level to another for a categorical moderator), SE of β, p value and number of effect sizes (k) included in each meta-regression model. Finally, study scores in the 'Checklist of issues that should be included in reports of studies of football injuries' were used as a continuous moderator to assess whether studies with lower reporting quality influenced estimates. Beta coefficient for reporting quality represents change in log (for incidence and burden) or arcsine transformed (for proportion and prevalence) outcome per 1 unit increase in quality.[33]

Equity, diversity and inclusion statement

The research team consists of one woman and four men from different disciplines with early, mid and late career researchers. We acknowledge the majority of authors are from one country and lack the perspectives of persons from middle-income to low-income countries. We included all eligible studies regardless of participants' sex, race/ethnicity/culture and socioeconomic level.

RESULTS

Search strategy

The electronic searches retrieved 1841 references from the literature. After duplicate removal and title and abstract screening, 227 full-text articles (81% from databases) were obtained and assessed for inclusion, resulting in 71 articles in the quantitative synthesis. The search results are presented in figure 1.

Assessment of reporting quality

Across all included studies, there was an average rating of 15 out of 19 (min=9, max=19). Full details of the reporting quality assessment for all included studies are provided in online supplemental table S3.

Description of studies

Participants

Across 71 studies,[2 3 5 6 20 36–101] a total of 9 441 381 hours of exposure were registered in 10 sports (k=193). 60 studies (85%) were conducted on a soccer population, two studies were conducted on rugby, field hockey, futsal and ice hockey (3% each), while one study was conducted on skiing, artistic gymnastics, basketball, tennis and gaelic football (1.4% each). There were 131 (68%) effect sizes presenting male data exclusively, 13 (6.7%) included mixed cohorts and 49 (25%) included exclusively female athletes. The median year of injury data collection was 2011; the earliest study recorded groin injuries in 1996, while the most recent studies collected data in 2022. Elite athletes represented 70% of the cohorts (n=52) while the rest were amateur athletes (n=21). From studies that reported anthropometric characteristics, the average age, body mass and height of the included participants was 22 years (min=12, max=28), 71 kg (min=45, max=100) and 1.75 m (min=1.48, max=1.85), respectively. Individual characteristics of the included studies are summarised in online supplemental table S4.

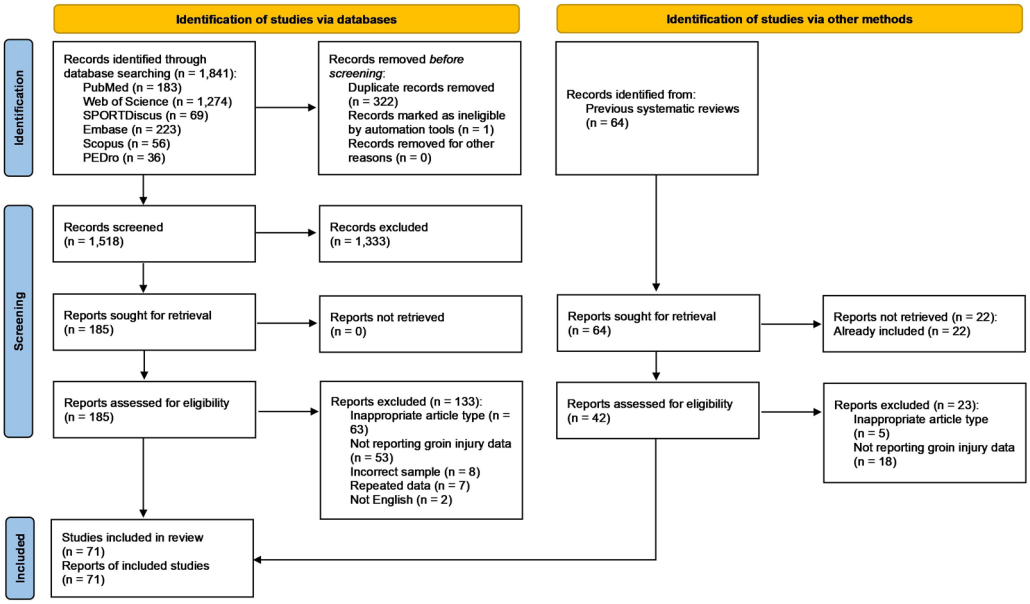

Figure 1 PRISMA flow diagram. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

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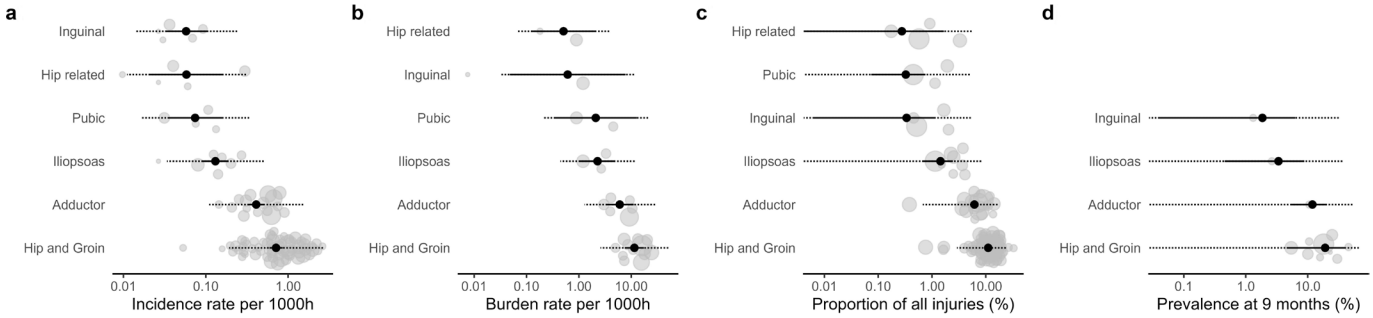

Figure 2 Forest plots for hip and groin injury meta-analysis. Panels show pooled estimates (black circles) with 95% CI and 95% PI, along with individual cohort values (grey bubbles) for different epidemiological measures: (a) incidence, (b) injury burden rate, (c) proportion of hip and groin injuries and (d) prevalence. Bubbles representing individual cohorts are sized according to their precision (ie, inversely to their SE). PI, prediction interval.

Injury definitions

Studies reported a total of 127 cohorts with HGI. Among these, adductor-related injuries were the most reported (n=35), followed by iliopsoas-related (n=14), hip-related (n=7), inguinal-related (n=6) and pubic-related injuries (n=4). There were 181 reports of time-loss injuries, while the 'all complaints' injury definition was presented 9 times, one study used a medical attention definition. The data on injury scenarios included total injuries (n=129), training injuries (n=30) and match injuries (n=34).

Quantitative synthesis of injury rates

Injury incidence

The overall HGI incidence rate was 0.71 per 1000 exposure hours (95% CI 0.59 to 0.86, PI=0.19 to 2.65; k=98, injuries=5914, total exposure=6 883 650 hours, figure 2a). Regarding data by clinical entities, adductor-related injury incidence was 0.41 per 1000 hours (95% CI 0.33 to 0.5, PI=0.11 to 1.53; injuries=3209, total exposure=5 707 416, figure 2a), iliopsoas-related injury incidence was 0.13 per 1000 hours (95% CI 0.09 to 0.19, PI=0.03 to 0.51; injuries=266, total exposure=2 562 659, figure 2a), pubic-related injury incidence was 0.07 per 1000 hours (95% CI 0.03 to 0.16, PI=0.02 to 0.34; injuries=89, total exposure=2 125 314, figure 2a), inguinal-related injury incidence was 0.06 per 1000 hours (95% CI 0.03 to 0.1, PI=0.01 to 0.24; injuries=98, total exposure=2 270 071, figure 2a) and hip-related injury incidence was 0.06 per 1000 hours (95% CI 0.02 to 0.16, PI=0.01 to 0.31; injuries=133, total exposure=2 303 168, figure 2a).

Injury burden

The overall injury burden rate for HGI was 11.5 days lost per 1000 hours of exposure (95% CI 7.53 to 17.49, PI=2.58 to 50.99; k=27, days lost=3041, total exposure=3 217 894, figure 2b). Injury burden was 6.02 (95% CI 3.27 to 11.11, PI=1.27 to 28.56; days lost=1325, total exposure=2 289 233, figure 2b) for adductor-related injuries, 2.27 (95% CI 1.04 to 4.97, PI=0.44 to 11.59; days lost=182, total exposure=1 997 039, figure 2b) for iliopsoas-related injuries, 2.1 (95% CI 0.35 to 12.74, PI=0.21 to 21; days lost=62, total exposure=1 882 305, figure 2b) for pubic-related injuries, 0.61 (95% CI 0.05 to 7.81, PI=0.03 to 11.36; days lost=68, total exposure=1 882 305, figure 2b) for inguinal-related injuries, and 0.51 (95% CI 0.12 to 2.09, PI=0.07 to 3.8; days lost=77, total exposure=1 882 305, figure 2b) for hip-related injuries.

Proportion of groin injuries relative to all injuries

HGIs represented 11% of all injuries (95% CI 10% to 13%, PI=3% to 24%; k=106, HGIs=5708, all injuries=46 959, figure 2c), adductor-related injuries represented 6% (95% CI 5% to 7%, PI=1% to 17%; adductor-related injuries=3215, all injuries=44 552, figure 2c), iliopsoas-related injuries represented 1% (95% CI 1% to 2%, PI=0% to 8%; iliopsoas-related injuries=270, all injuries=17 127, figure 2c), inguinal, pubic and hip-related injuries represented <1% of all injuries (figure 2c). All results regarding the proportion of groin injuries relative to all injuries are presented in online supplemental table S5.

Injury prevalence

Seasonal (9-month season) HGI prevalence was 19% (95% CI 5% to 40%, PI=0% to 67%; k=12, injured players=1883, total population=10 748, figure 2d), prevalence of adductor-related injuries was 12% (95% CI 5% to 20%, PI=0% to 53%, injured players=16, total population=152, figure 2d), prevalence of iliopsoas-related injuries was 3% (95% CI 0% to 9%, PI=0% to 36%, injured players=4, total population=152, figure 2d) and prevalence of inguinal-related injuries was 2% (95% CI 0% to 6%, PI=0% to 32%, injured players=2, total population=152, figure 2d). It was not possible to calculate injury prevalence for pubic-related nor hip-related injuries due to insufficient data available.

Small study bias

Evidence of small study bias was confirmed by the modified Egger's regression test and visually inspecting funnel plots for groin injury incidence and proportion of groin injuries relative to all injuries (online supplemental figure S1). Results from the sensitivity analyses are presented in online supplemental tables S6–7.

Meta-regression

Moderator analysis of continuous variables revealed that the incidence of HGI increased with higher age (+5.12% for every 1-year increase, β=0.05, SE=0.03, p=0.046, k=53, figure 3a), body mass (+3% for every 1 kg increase, β=0.03, SE=0.01, p=0.004, k=40, figure 3b), height (+2.8% for every 1 cm increase, β=2.62, SE=1.12, p=0.024, k=40, figure 3c) and higher proportion of male athletes studied (β=−0.6, SE=0.2, p=0.004, k=61, figure 3f). There was no effect of BMI (+11%, β=0.11, SE=0.06, p=0.052, k=40, figure 3d), or year of data collection (+2%, β=0.02, SE=0.01, p=0.127, k=61, figure 3e)

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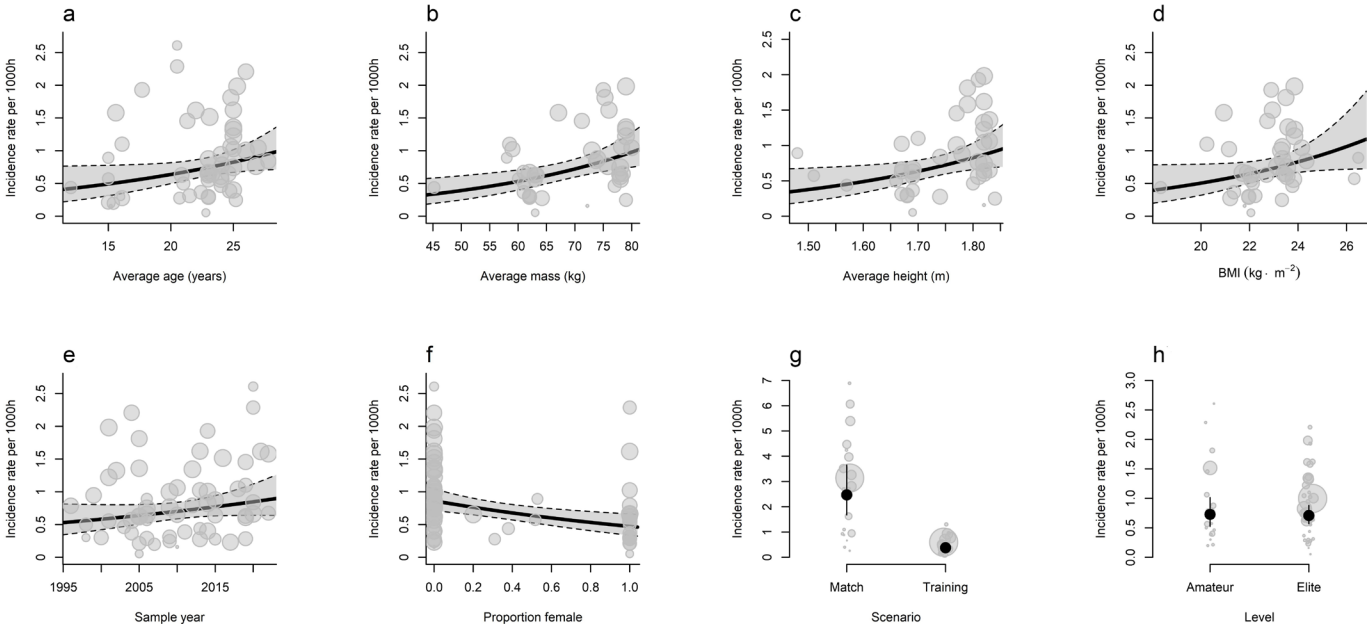

Figure 3 Moderator (meta-regression) analysis for hip and groin injury incidence. Panels show pooled incidence (black line) with 95% CI (grey shaded) and individual cohort incidence (grey bubbles) against various moderators: (a) average age, (b) body mass, (c) height, (d) BMI, (e) median year of observation, (f) proportion of female athletes, (g) match versus training and (h) amateur versus elite athletes. Bubbles representing individual cohorts are sized according to their precision (ie, inversely to their SE). BMI, body mass index.

on the estimates. Sensitivity analysis revealed no effect of age-adjusted average height (−1%, β=−0.01, SE=0.04, p=0.87, k=40, (online supplemental figure S2) and body mass (−2.9%, β=−0.03, SE=0.04, p=0.52, k=40, online supplemental figure S2). Details of the different meta-regression models are presented in online supplemental table S8.

Regarding the categorical moderators, injury incidence was significantly higher (β=−1.88, SE=0.15, p<0.001, figure 3g) in competitions (incidence rate=2.47, 95% CI 1.67 to 3.65) compared with training sessions (incidence rate=0.38, 95% CI 0.27 to 0.53), while no significant differences (β=−0.03, SE=0.2, p=0.89, figure 3h) were noted between amateur (incidence rate=0.73, 95% CI 0.52 to 1.01) and elite athletes (incidence rate=0.71, 95% CI 0.57 to 0.88).

Sensitivity analyses

We conducted a sensitivity analysis to account for the potential influence of studies with lower reporting quality. Results were robust to reporting quality for all meta-analytical models of HGI (online supplemental table S9).

DISCUSSION

This is the first study to quantitatively assess the main epidemiological measures for HGIs in sport while considering the different clinical entities that can cause HGIs, including the largest groin injury dataset with 5914 injuries over 9 million exposure hours. These advancements enhance the accuracy and reliability of our estimates. The main findings were: (1) the incidence of HGIs was 0.71 per 1000 hours, with adductor-related injuries accounting for 0.41 per 1000 hours; (2) players missed an average of 11.5 days per 1000 hours due to HGIs; (3) HGIs comprised 11% of all injuries, while adductor-related injuries accounted for 6% and (4) the seasonal prevalence over a 9-month period was 19%. Secondary outcomes were a higher incidence in male athletes compared with females and increased incidence with age, while other anthropometric data did not influence the outcomes.

How common are HGIs in sport?

Our results suggest that athletes will suffer 0.71 injuries per 1000 hours of exposure, missing 11.5 days along that period due to injury. These results are robust due to the large number of injuries (n=71) captured and reflect a more confident estimate than the previous systematic review (n=34) on this topic, which reported an injury incidence ranging from 0.1 to 2.1 injuries per 1000 hours.[102] HGIs are among the most common injuries observed in sport in terms of injury incidence, comparable to hamstring (0.7 to 0.8/1000 hours) and quadriceps (0.5 to 0.8/1000 hours) injury rates,[47 103 104] and above the incidence of calf muscle strains (0.31/1000 hours).[47] Nonetheless, our findings and the studies mentioned above are primarily based on football populations. It is possible that incidence rates vary in other sports, so generalisation of these findings to other disciplines should be done with caution.

Analysing injury incidence while considering injury burden improves the understanding of which injuries necessitate further efforts to reduce injury rates. HGIs fall into the category of frequent and moderately burdensome injuries, with an average of 11.5 lay-off days per 1000 hours.[26] From the prevention standpoint, uncommon and severe (eg, anterior cruciate ligament; 30 lay-off days per 1000 hours), or frequent but less severe (eg, hamstring strains; 17.5 lay-off days per 1000 hours) injuries require more resources in order to reduce injury rates.[105] Our results confirm that preventing HGIs should take priority over other musculoskeletal problems in sport that have lower incidence and burden.

The role of diagnosing injuries according to clinical entities

One of the novelties presented in this systematic review is the quantitative synthesis of HGIs according to their specific

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clinical diagnosis. Adductor-related groin injuries represented by far the largest incidence (0.41/1000 hours) and burden (6/1000 hours) rates, with iliopsoas-related injuries representing 0.13/1000 hours and 2.3/1000 hours, respectively. It must be noted that the other clinical entities are less frequent and severe, with inguinal-related, pubic-related and hip-related groin injuries accounting for <1% of all injuries. Therefore, in designing and implementing prevention programmes, exercises should incorporate drills that require high electromyographic activity of the hip adductors,[106] given the importance of hip adduction strength in the development of groin injuries and the high injury rates for this specific clinical entity.[106 107]

From the clinical perspective, we should also consider the possibility of confronting patients with clinical entities that are less frequent (eg, inguinal-related or hip-related groin injuries) or multiple entities simultaneously, and avoid confirmation bias when assessing an athlete presenting with a groin injury.[108] For example, older athletes might suffer more hip-related groin injuries,[109] while younger athletes may be more prone to sustaining pubic-related injuries due to immature pubic symphysis.[13 83 110] That being said, since hip-related injuries are uncommon and of moderate severity in high-level athletes,[2] clinicians should be well-informed before performing interventional procedures (eg, FAI resection), as these may not be superior to conservative management.[111]

Are HGIs more common in male athletes?

The idea that males have a higher risk of sustaining groin injuries compared with female athletes is well widespread; the overall risk ratio is 2.45 in favour of males when men and women play the same sport.[22] Our data support this statement, since male athletes registered an overall injury incidence of 0.87/1000 hours, doubling the rates observed in females (0.47/1000 hours; incidence rate ratio=1.8). Differences in hip and pelvis morphology between sexes have been proposed as the factors that pose higher risk of injury in male athletes, although the exact causes are not well understood yet.[112] On the contrary, recent studies have found a high prevalence of groin injuries in female athletes. It is possible that groin injuries are underreported in females due to reduced access to medical staff, and that the recent professionalisation of female sport has brought resources that allow females to be better diagnosed.[76 113 114] With sport participation increasing in females, more studies are required to properly understand the problem of groin injuries in this population.

What other factors influence the appearance of HGIs in sport?

As part of our secondary aims, we investigated the effect of potential moderators on injury incidence. The average age of each cohort significantly moderated the incidence of HGI, with a 5% increase in injury incidence for every additional year of athlete age. The rest of the anthropometric variables did not yield significant effects once the average cohort age was accounted for. These results are in accordance with the current evidence around muscle injuries. Indeed, higher age has been associated with increased risk of injury for hamstring (+ 6.4%) and calf muscle groups,[103 115] while there is no clear link between age and quadriceps femoris strains.[116] The relationship between injury and age may be explained in two ways. Older athletes often have a history of previous injuries, which can lead to irreversible tissue changes like fatty infiltration and muscle atrophy, increasing their risk of reinjury.[117 118] Another possibility is that older athletes have diminished capacity for muscle regeneration, further raising the likelihood of injury.[119] Although adults face a higher risk of sustaining HGI compared with adolescents, the latter still require careful monitoring. Young athletes are particularly vulnerable to groin injuries during growth spurts due to the combination of a rapid increase in adductor strength and the appearance of abnormal movement patterns, placing additional stress on the groin area.[120 121] To reduce the rates of HGI, the authors advise tracking groin symptoms in players of all ages, given the susceptibility to injury during both adolescence and adulthood.

Clinical implications

Based on our findings, routine screening and ongoing monitoring of groin symptoms should be a core component of athlete health surveillance to identify problems early and prevent progression to time-loss injuries. Regarding prevention, practitioners are advised to prioritise exercises targeting the hip adductors, as this review identifies adductor-related injuries as the most frequent and burdensome clinical entity. Furthermore, the positive association between age and injury incidence suggests that management strategies must be age-specific: older athletes warrant tailored load management to mitigate recurrence risk, while younger players require close monitoring during growth spurts.

LIMITATIONS

First, due to the low number of studies conducted outside a soccer environment (15%), no subanalysis could be performed to compare incidence rates by sport. Thus, estimates may be soccer-biased due to the weight of this sport in the meta-analytical models. With the currently available evidence, the reader should consider this when interpreting the results. Addressing this gap requires more research in less represented sports to allow for meaningful between-sport injury rate comparisons. Second, terminology around HGIs was not well established until 2015 and various studies included in the review were not following the definitions proposed by the Doha agreement, which could have introduced bias in the overall and clinical entity estimates.[18] For instance, a study may have used the definition of 'adductor tendinitis' while collecting all episodes of groin injuries, regardless of whether they affected the adductor muscles. Third, we required studies to follow a prospective design to be included. This decision may have led to reports not included due to having a cross-sectional or retrospective design, while they could still be valuable for reporting epidemiological data. In this regard, many cross-sectional studies report seasonal prevalence rates, while only nine studies (k=12) included in our analysis presented results on injury prevalence. Fourth, groin injuries comprise cases that lead to activity cessation but also injuries that diminish performance without time loss, although the trend on reporting non-time-loss data has raised in the last years; most of the included articles only reported time-loss cases. Considering time-loss injuries exclusively leads to an underestimation of injury rates, since up to 90% of injuries in some cohorts do not require lay-off periods.[20] Some characteristics (eg, athlete sex) may influence the occurrence of non-time-loss groin problems,[122] which may have been overlooked with the applied analysis. Fifth, included studies did not always report sufficient data to maximise their use in our subanalyses. For example, data across various seasons

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was reported as a single value, and certain studies did not present demographic data to include as moderators. We were able to perform influence analyses using anthropometric data alongside scenario and competitive level, although it was not possible to perform meta-regression using other groin injury risk factors (eg, adductor strength) due to insufficient reporting in the included studies. Sixth, no MeSH terms were used in the literature search, and their omission may have increased the risk of missing some relevant studies. Seventh, no formal risk of bias assessment was performed, and consequently, a GRADE (Grading of Recommendations, Assessment, Development and Evaluation) assessment of the quality of evidence could not be conducted. Finally, one experienced author extracted data from the included studies, which may have introduced errors in the collected information. According to the Cochrane Handbook, it is highly desirable that at least two authors independently extract data, particularly outcome data, to minimise errors and reduce potential bias.

CONCLUSIONS

The incidence of HGIs is 0.7 per 1000 hours of exposure, comprising 11% of all sporting injuries. Stakeholders can expect players to miss 12 days for every 1000 exposure hours due to HGIs. Incidence rates increase with higher age, while males are at higher risk of injury compared with females. Neither anthropometric characteristics nor elite status appears to influence incidence rates, and notably, these rates have not declined over the past 30 years. Our findings highlight the need for further efforts to reduce the occurrence of HGIs in sports.

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